

Clinical Outcomes of Varicocele Repair in Infertile Men: A Review.

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Varicoceles are a major cause of impaired spermatogenesis and the most common correctable cause of male infertility. They are found in approximately 40% of men with primary infertility and 80% of men with secondary infertility, although they also occur in 12% of men with normal semen parameters. The presence of a varicocele does not always affect spermatogenesis, as it has been reported that only 20% of men with documented varicoceles suffer fertility problems. However, varicocele repair appears to have beneficial effects in men with impaired semen parameters and palpable varicoceles. Currently, the main procedures employed for varicocele repair are microsurgical subinguinal or inguinal varicocelectomy, laparoscopic varicocelectomy, and radiological percutaneous embolization. Microsurgical varicocelectomy appears to be the optimal treatment in most cases, whereas the other procedures are useful only in specific cases. After treatment, it typically takes 3 to 6 months for patients' semen parameters to improve; thus, other therapies, including assisted reproductive technology, should be considered if infertility persists after this interval, especially in older couples. Controversies still remain regarding how varicoceles in certain subgroups, such as adolescents or men with azoospermia, should be treated. Due to their relatively high prevalence rate among the general population, varicoceles can occur concomitantly with other conditions that cause impaired spermatogenesis. Further studies are necessary in order to identify the patients who are most likely to benefit from treatment. In this review, we sought to summarize the issues currently associated with varicocele treatment in infertile men.

[Needle aspiration testicular cytology as diagnostic parameter in the assessment with varicocele].

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Varicocele is considered the most common cause of male infertility. The pathogenesis of the spermatogenetic impairment in this disease is not completely understood and the seminal patterns of patients with varicocele are different. Testicular fine-needle aspiration cytology (FNAC) in infertile men allows us to obtain many information about the type and the degree of alteration of the seminiferous tubule. In this study, 200 infertile men with varicocele were evaluated for seminal parameters, FSH, LH and testosterone plasma concentrations. Testicular pattern was analyzed by bilateral FNAC. All patients suffered of infertility for at least two years and varicocele was the sole infertility-related pathology. According to the tubular patterns, we divided the patients into five groups, and we observed that the different tubular alterations could be present in patients with similar FSH levels and similar sperm count. Then, to verify the role of tubular pattern as prognostic factor for the correction of varicocele, we studied 70 men, performing seminal analysis and hormonal evaluation six months after sclerotherapy. We found an increasing in sperm count and in other seminal parameters after the sclerotherapy (motility, normal morphology...) only in patients with maturative alteration at spermatid level. This alteration of spermiogenesis seems to be reversible after sclerotherapy. In the light of these results the testicular FNAC gives important prognostic parameters in the selection of patients affected by varicocele for sclerotherapy.

The benefits of varicocele repair for achieving pregnancy in male infertility: A

systematic review and meta-analysis.

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Varicocele is one of the most common treatable causes of male infertility. However, the decision to perform varicocelectomy before starting a fertility program remains controversial. This study aimed to thoroughly review and analyze the benefit of varicocele repair and its impact on the success rate of a fertility program. A systematic literature search was performed using MEDLINE, Cochrane Library, and Wiley Library. The primary outcome was the pregnancy rate, and the secondary outcomes were live birth rate and surgical sperm retrieval success rate. Outcomes were compared between men who underwent treatment for a varicocele and those that did not. The pooled analysis data are presented as odds ratios with 95% confidence intervals. A total of 31 articles were included in the meta-analysis. The pregnancy rate was significantly higher in the treated group (odds ratio = 1.82; 95% confidence interval: 1.37-2.41; $P < 0.0001$) along with the live birth rate (odds ratio = 2.80; 95% confidence interval: 1.67-4.72; $P = 0.0001$). The further subgroup analysis revealed a higher pregnancy rate in treated men with azoospermia, subnormal semen parameters, and normozoospermia ($P = 0.04$, $P = 0.0005$, and $P = 0.002$, respectively), while the live birth rate was only significantly higher in the treated men with subnormal semen parameters and normozoospermia ($P = 0.001$ and $P < 0.0001$). Treated varicocele also led to a higher sperm retrieval rate in azoospermic patients (odds ratio = 1.69; 95% confidence interval: 1.16-2.45; $P = 0.006$). Varicocele repair increased the pregnancy and live birth rates regardless the semen analysis result, along with the sperm retrieval success rate in azoospermic men. Thus, varicocele repair may be beneficial prior to joining a fertility program.

Revisiting the impact of varicocele and its treatments on male fertility.

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Varicocele is one of the most common, yet treatable, causes of male infertility. Varicoceles are present in more than 40% of infertile men with primary infertility, a figure that increases with age. Varicoceles impair semen parameters and sperm DNA and are linked with lower pregnancy and live birth rates. Until recently, men had seldom been examined in male fertility workups. This is changing as urologists have become recognized as team members in infertility. Hence identification and treatment are available as never before. Furthermore, as men become aware that they are as likely as their female partners to be infertile, they want equal 'couple care', requesting urological referrals as they realize that they can improve their semen quality and chances of fatherhood without or before fertility treatment. There is now a greater understanding of the mechanisms of varicocele-induced damage by oxidative stress, using sperm DNA as a sensitive biomarker of sperm quality. There is a current consensus that varicocele is linked to poor semen and repair improves semen and sperm DNA quality. Evidence is strengthening to indicate that varicocele repair increases pregnancy and live birth rates in natural conception and following fertility treatment.

Testicular arterial blood flow volume in predicting semen improvement following microscopic subinguinal varicocelectomy.

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Varicocele is a significant but treatable contributor to male infertility. The efficacy of varicocelectomy in improving sperm quality is not consistent, with only 60-80% of patients experiencing improved semen quality. This prospective cohort study aimed to evaluate the effect of microscopic subinguinal varicocelectomy (MSV) on testicular arterial blood flow volume (TABFV) and to determine the value of preoperative TABFV in predicting the outcome of MSV. Patients with varicocele who underwent MSV at the same clinical center between July 2020 and April 2023 were enrolled. All patients underwent ultrasound assessment and at least one semen analysis before and after MSV. Both univariate and multivariate logistic regression analyses were performed to assess the association between pre-MSV variables and semen improvement after MSV. Subsequently, a diagnostic model was developed. This study enrolled 96 patients with varicocele, including 31 who showed semen improvement after MSV and 65 who did not. The postoperative semen-improved group demonstrated a significant increase in TABFV of the right testis (TABFV-R) and left testis (TABFV-L) ($P < 0.001$). Notably, the postoperative TABFV-L was more than twice the preoperative TABFV-L. Preoperative TABFV-R and a combination of subclinical right-sided varicocele were found to be associated with semen improvement after MSV, and a diagnostic model was developed using these two variables. The diagnostic model exhibited satisfactory performance, with an area under the curve (AUC) of 0.824 [95% confidence interval (CI): 0.735-0.913], which was further validated internally yielding an AUC of 0.824 (95% CI: 0.726-0.900). Additionally, calibration analysis confirmed that the diagnostic model was well calibrated, and the Hosmer-Lemeshow test resulted in a P value of 0.794. The decision curve demonstrated that using this proposed nomogram would yield a net benefit if the threshold probability for semen improvement after MSV exceeded 10%. TABFV-L demonstrated potential utility in clinical practice for assessing outcomes of MSV, and the diagnostic model incorporating TABFV-R and a combination of right-side varicocele performed well in predicting improvements in semen parameters following MSV.

Research advances in inflammation and oxidative stress in varicocele-induced male infertility: a narrative review.

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Varicocele, the most common and treatable cause of male infertility, significantly impacts fertility. The pathophysiological mechanisms of varicocele have not been fully understood yet. Recent studies have focused on the pathophysiology of varicocele-induced infertility, highlighting inflammation and oxidative stress as key contributing factors. We reviewed recent research on the roles of inflammation and oxidative stress in the pathophysiology of varicocele and found that they negatively impact semen parameters, spermatogenesis, and testicular and epididymal function. In addition, this article summarizes the related factors of inflammation and oxidative stress caused by varicocele. Finally, a brief consideration on the treatments to address inflammation and oxidative stress is proposed. This review may provide treatment options and targets for varicocele-induced infertility. However, the relationship between inflammation and oxidative stress in varicocele still needs further study.

Does Sperm DNA Fragmentation Index Continuously Decrease Over Time After Varicocelectomy in Varicocele-Induced Infertility? A Systematic Review and Meta-Analysis.

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Varicocele (VC) is the most frequent and reversible cause of male infertility. One of the preferred management strategies to alleviate this problem is varicocelectomy. However, there are no researchers who have explored the relationship between better timing and postoperative sperm DNA fragmentation index (DFI) improvement in patients. We conducted this meta-analysis by enrolling published studies to find out the best waiting time after varicocelectomy to wait for better improvement of sperm DFI. A literature search was conducted using PubMed, Embase, Scopus, Web of Science, and Cochrane Library databases. The data from the pooled analysis were presented as mean difference (MD) along with a 95% confidence interval (CI). Heterogeneity was evaluated using I². Four studies were included after screening relevant literature. Statistical analysis revealed that after varicocelectomy, follow-up results within 3 months showed a significant improvement in sperm DFI compared with the preoperative period (MD: -3.66, 95% CI = [-5.17, -2.14], $p < .00001$), and follow-up results with 6 months showed a significant improvement in sperm DFI compared with the postoperative 3 months as well (MD: -1.51, 95% CI = [-2.73, -0.29], $p = .02$). Notably, no further improvement in sperm DFI was observed when the follow-up period reached 12 months (MD: -1.59, 95% CI = [-3.22, 0.05], $p = .06$). Six months after varicocelectomy may be the optimal time for sperm DFI compared with 12 months or even longer, which means it is also the preferable time for conception. However, more well-designed prospective studies are needed in the future to validate our conclusion.

Machine learning-based classification of varicocoele grading: A promising approach for diagnosis and treatment optimization.

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Varicocoele is a correctable cause of male infertility. Although physical examination is still being used in diagnosis and grading, it gives conflicting results when compared to ultrasonography-based varicocoele grading. We aimed to develop a multi-class machine learning model for the grading of varicocoeles based on ultrasonographic measurements. Between January and May 2024, we enrolled unilateral varicocoele patients at an infertility clinic, assessing their varicocoele stages using the Dubin and Amelar system. We measured vascular diameter and reflux time at the testicular apex and the subinguinal region ultrasonography in both the supine and standing positions. Using these measurements, we developed four multi-class machine learning models, evaluating their performance metrics and determining which patient position and projection were most influential in varicocoele grading. We included 248 patients with unilateral varicocoele in the study, their average age was 26.61 ± 4.95 years old. Of these, 212 had left-sided and 36 had right-sided varicocoeles. According to the Dubin and Amelar system, there were 66 grade I, 96 grade II, and 86 grade III varicocoeles. Among the models we created, the random forest (RF) model performed best, with an overall accuracy of 0.81 ± 0.06 , an F1 score of 0.79 ± 0.02 , a sensitivity of 0.69 ± 0.02 , and a specificity of 0.8 ± 0.03 . Vascular diameter measurement at the testicular apex in the supine position had the most impact on grading across all models. In support vector machine and multi-layer perceptron models, reflux time measurements from the subinguinal projection in the standing position contributed the most, while in RF and k-nearest neighbors models, measurements from the subinguinal projection in the supine position were the most influential. Machine learning methods have demonstrated superior accuracy in predicting disease compared to traditional statistical regressions and nomograms. These advancements hold promise for clinically automated prediction of varicocoele grades in patients. Tailored varicocoele grading for individuals has the potential to enhance treatment effectiveness and overall quality of life.

Pathophysiology and treatment options of varicocele: An overview.

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Varicocele is the most common correctable cause for male infertility, but not all men with varicocele are affected equally by this condition. The pathophysiology of varicocele-induced fertility remains ill-defined. While evidence suggests that oxidative stress remains a central factor, other mechanisms likely include scrotal hyperthermia, reflux of metabolites, hypoxia and cadmium accumulation. Microsurgical varicocelectomy remains the gold standard treatment option for infertile men with a clinically palpable varicocele and abnormal semen parameters. Newer evidence suggests a potential role for antioxidant supplementation and a meaningful role of varicocelectomy for patients destined for ART to improve pregnancy outcomes.

Ultrasound evaluation of varicoceles: guidelines and recommendations of the European Society of Urogenital Radiology Scrotal and Penile Imaging Working Group (ESUR-SPIWG) for detection, classification, and grading.

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Varicoceles are relatively common particularly in asymptomatic men and are even more prevalent in subfertile men, representing the most common potentially correctable cause of male infertility. Ultrasound (US) is the imaging modality of choice for varicocele evaluation, but there is no widely accepted consensus on examination technique, diagnostic criteria, or classification. In view of this uncertainty, the guideline writing group (WG) of the European Society of Urogenital Radiology (ESUR) Scrotal and Penile Imaging Working Group (ESUR-SPIWG) undertook a literature review and assessment of the quality of relevant evidence. The group then produced evidence-based recommendations for varicocele US examination, interpretation, and classification by consensus agreement. The results are presented in the form of 15 clinical questions with a brief summary of the relevant evidence and the authorised recommendations from the SPIWG. This paper provides a short summary of the evidence evaluation and the complete recommendations.

Key Points

- Varicocele is a common clinical problem; it is highly prevalent amongst subfertile men and the most common potentially correctable cause of male infertility.
- Ultrasound is the imaging modality of choice for varicocele assessment, but there is no generally agreed consensus on the US examination technique or the criteria that should be used for diagnosis, grading, and classification.
- This paper summarises the recommendations of the ESUR-SPIWG for standardising the US assessment of varicoceles. This includes examination technique, image interpretation, classification, and reporting.